

Rational Choice Overload

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Introduction

Theory

- Standard Sequential Search

- Overview

- Model Representation

- Optimal Search

Experiment

Results

Introduction: what's choice overload?

- ▶ Choice overload:
 - “the paradoxical finding that providing individuals with more options can be detrimental to choice” (Chernev et al., 2015)
- ▶ More heavily studied in Psychology and Marketing, more lightly in Economics

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- ▶ Detrimental? Various possible definitions.

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 - Others very much within it: choice

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- ▶ More heavily studied in Psychology and Marketing, more lightly in Economics
- ▶ Detrimental? Various possible definitions.
 - Some outside traditional scope of Economics: confidence*, satisfaction
 - Others very much within it: choice
- ▶ This project: default choice
 - not choosing one of the options presented by staying with a default choice
 - choice overload: more default choice in larger choice sets

Introduction: preview of this paper

- ▶ Previous explanations rely on: psychological assumptions or strategic interaction
 - Psychological assumptions (decision avoidance)
 - Strategic Interaction (contextual inference)
 - choice overload happens because people don't engage with large choice sets

Introduction: preview of this paper

- ▶ We show choice overload can result from rational optimal behavior within search models
 - with mix of theory and an experiment
 - people start engaging even with large choice sets, but might stop before finding options they like
 - choice set composition matters!

Introduction: origins

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- ▶ Spark: Iyengar & Lepper(2000)
- ▶ Display of jams in a local supermarket
 - 6 jams vs. 24 jams (superset)
- ▶ Far more people buy jam in 6 jams treatment: 12% vs. 2%

Introduction: a troubled literature

- ▶ Literature on choice overload has stood in a state of trouble and inconclusiveness.

Introduction: a troubled literature

- ▶ Literature on choice overload has stood in a state of trouble and inconclusiveness.
- ▶ Attempts at directly replicating the original results were unsuccessful (Scheibehenne et al., 2010, p.412).
 - although similar findings in independent studies have been reported (Chernev et al., 2015, p. 335)
- ▶ Some meta-analyses, like Scheibehenne et al. (2010), conclude in favor of the non-existence of choice overload.
- ▶ Chernev et al. (2010), on the other hand, criticize their methods and conclusions.

Introduction: a troubled literature

- ▶ Dean, Ravindran and Stoye (WP) provide better tests and more solid evidence for CO
- ▶ Yet, no theoretical explanation has solidified itself as standard
- ▶ The puzzle of choice overload: missed good alternatives

Introduction: the interest of choice overload

Theoretically:

- ▶ Choice overload yet another violation of IIA
- ▶ But also violation of, e.g., monotonicity in random utility models

Practically:

- ▶ Implications for design of choice environments
 - important markets such as retirement plans and health insurance
- ▶ Our approach: different recommendations than existing models
- ▶ Before: people don't engage with large choice sets
- ▶ Our approach: people do, but might stop engaging before finding something they like

Introduction: this paper

- ▶ This paper: choice process leading to choice overload, even under optimality?

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- ▶ Choice process: boundedly rational sequential information sampling

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 - Sequential search framework.

Introduction: this paper

- ▶ This paper: choice process leading to choice overload, even under optimality?
- ▶ Choice process: boundedly rational sequential information sampling
 - Sequential search framework.
- ▶ Goal: develop a family of models that can explain choice overload using this search framework, and test them using a laboratory experiment.
- ▶ Explore *non-constant thresholds*:
 - Period dependent.
 - History dependent

Introduction: Other approaches

Currently two main classes of model used to explain choice overload:

- ▶ **Contextual strategic inference.** Kamenica (2008), Kuksov and Villas-Boas (2010), Nocke and Rey (2021)
 - DM can make inferences about the nature of a set of alternatives based on its size, and the assumption that the set has been chosen by a profit maximizing firm.
- ▶ **Decision avoidance.** Beattie et al. (1994), Dean (2008), Gerasimou (2018)
 - DM avoids engaging with large choice sets because they find it aversive to do so.

Call these two explanations “classic” choice overload.

Basic Overview

- ▶ Explore how sequential search models can generate choice overload.
- ▶ Standard search model cannot, though.
- ▶ Introduce extensions of the model which can, each with different behavioral predictions.
- ▶ Experiment that allows us to differentiate between these models, and also between existing explanations.

Introduction

Theory

- Standard Sequential Search

- Overview

- Model Representation

- Optimal Search

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Preliminary Notes

- ▶ Instead of going over all formal results, will present the *logic* of the theory section.
- ▶ Present a synoptic view of what the section is doing, and how the pieces interact together

A Refresher: Standard Sequential Search

Standard sequential search framework:

- ▶ Decision maker faces finite set of alternatives X
- ▶ Has to sequentially search through X
- ▶ There's a *default option* $d \in X$, which starts out searched.

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- ▶ Decision maker faces finite set of alternatives X
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- ▶ There's a *default option* $d \in X$, which starts out searched.
- ▶ Sequentially searching through X :
 - In each period, can *stop* and pick the *best option seen so far* y , or *search* at cost k and look at a new option x drawn from a known distribution F .

A Refresher: Standard Sequential Search

Three points:

A Refresher: Standard Sequential Search

Three points:

- ▶ Differences to *job search*: recall, finite horizon, no arrival process

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- ▶ Costs
 - Theory agnostic about what costs are exactly
 - For our applications: cognitive costs

A Refresher: Standard Sequential Search

Three points:

- ▶ Differences to *job search*: recall, finite horizon, no arrival process
- ▶ Costs
 - Theory agnostic about what costs are exactly
 - For our applications: cognitive costs
- ▶ Options are iid draws from a specific distribution (from DM's perspective)
 - Ex ante identical
 - Search order irrelevant for optimality
 - Search order controlled in experiment

A Refresher: Standard Sequential Search

Bellman equation for the standard model:

$$V_n(y_n) = \max \left\{ u(y_n), -k + \int V_{n+1}(y_{n+1}(x)) dF(x) \right\}$$

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Solution: constant threshold τ over utility values such that:

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- ▶ searches otherwise

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- ▶ DM stops if $u(y) \geq \tau$
- ▶ searches otherwise

τ doesn't change during search, nor does it change with choice set size

A Refresher: Standard Sequential Search

- ▶ Such constant thresholds cannot lead to choice overload.
Why?

A Refresher: Standard Sequential Search

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Why?

- If d not chosen in a small set, search started, i.e. $u(d) < \tau$,
and something better got chosen

A Refresher: Standard Sequential Search

- Such constant thresholds cannot lead to choice overload.

Why?

- If d not chosen in a small set, search started, i.e. $u(d) < \tau$, and something better got chosen
- Thus search also starts in the larger set, continues until better option is found

A Refresher: Standard Sequential Search

$$V_n(y_n) = \max \left\{ u(y_n), -\textcolor{red}{k} + \int V_{n+1}(y_{n+1}(x)) d\textcolor{blue}{F}(x) \right\}$$

Making changes to:

- ▶ Cost function.
- ▶ (Beliefs about the) distribution of values.

Leads to non-constant thresholds:

- ▶ History dependent: what options have been seen.
- ▶ Period dependent: number of options searched.

Theory Overview

Data

Threshold Functions

Optimal Behavior

Theory Overview

Data

Threshold Functions

Optimal Behavior

Models of optimizing agent

Theory Overview

Data

Description of observed behavior

Threshold Functions

Optimal Behavior

Models of optimizing agent

Theory Overview

Data

Description of observed behavior

Threshold Functions

Representing solutions from optimal models

Optimal Behavior

Models of optimizing agent

Theory Overview

Data

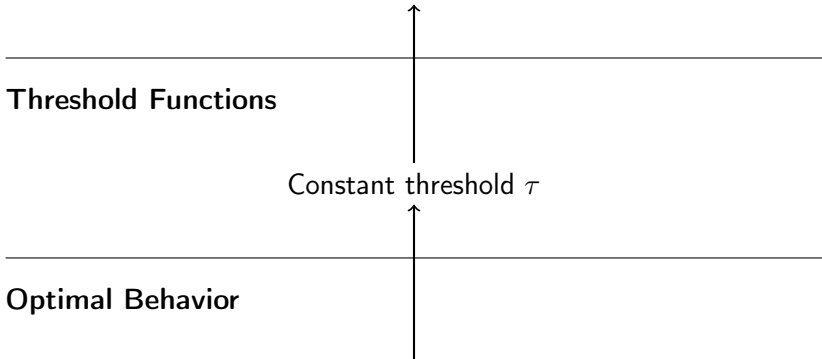
Data that cannot exhibit choice overload

Threshold Functions

Constant threshold τ

Optimal Behavior

Standard sequential search



Theory Overview

Data

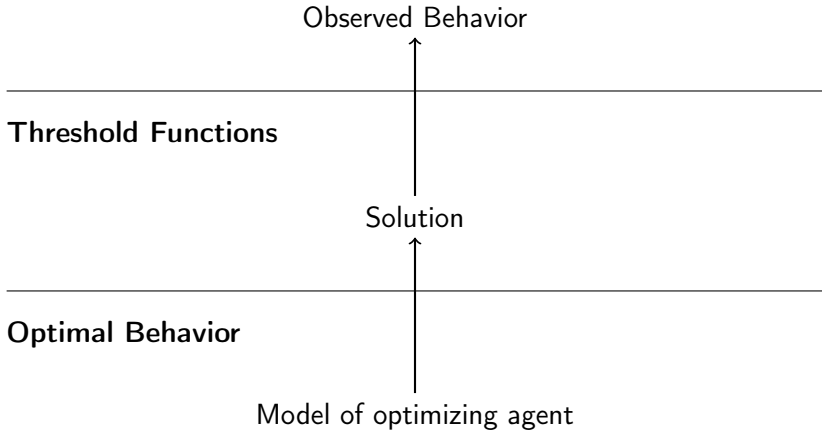
Observed Behavior

Threshold Functions

Solution

Optimal Behavior

Model of optimizing agent



Top Layer: Data

Basics:

- ▶ Finite *grand set* of alternatives X
- ▶ $d \in X$ is *default option*
- ▶ $\mathcal{A} = \{A \in 2^X \mid d \in A\}$
- ▶ $\mathcal{L}_A$: set of linear orders on A starting with d
- ▶ $\mathcal{L} = \cup_{A \in \mathcal{A}} \mathcal{L}_A$
 - $I \in \mathcal{L}$ is a *choice list*

Top Layer: Data

We consider two types of data

1. Standard stochastic choice data: $p(a, A)$
 - A is a choice set (always contains d)
 - $p(a, A)$ is the probability of choosing a from set A
 - Allows us to determine which models generate choice overload

Top Layer: Data

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2. Experimental choice data: $\rho(x, y, l)$

- l is a choice list (ordered set of elements in X , with d the first element)
- Probability that search stops at y and x is chosen from list l
 - ▶ x as chosen option, choice point
 - ▶ y as stopped option, stopping point
- Allows us to differentiate between causes of choice overload
- Will collect such data in our experiment

Top Layer: Data

- Given standard choice data: $p(a, A)$:

Definition

We say that a data set **exhibits choice overload** if, for some A, B such that $A \subset B$, $p(d, B) > p(d, A)$.

Top Layer: Data

Connecting $\rho(x, y, l)$ to $p(x, A)$:

Top Layer: Data

Connecting $\rho(x, y, l)$ to $p(x, A)$:

$$\rho(x, y, l)$$

Top Layer: Data

Connecting $\rho(x, y, l)$ to $p(x, A)$:

$$\sum_{y \in A} \rho(x, y, l)$$

- Aggregate across possible stopping points

Top Layer: Data

Connecting $\rho(x, y, l)$ to $p(x, A)$:

$$\pi_A(l) \sum_{y \in A} \rho(x, y, l)$$

- ▶ Aggregate across possible stopping points
- ▶ Pick a distribution over orderings of A : π_A

Top Layer: Data

Connecting $\rho(x, y, l)$ to $p(x, A)$:

$$p(x, A) = \sum_{l \in \mathcal{L}_A} \pi_A(l) \sum_{y \in A} \rho(x, y, l)$$

- ▶ Aggregate across possible stopping points
- ▶ Pick a distribution over orderings of A : π_A
- ▶ Aggregate across orders

Theory Overview

Data

$$p(a, A), \rho(x, y, l)$$

Threshold Functions

Solution

Optimal Behavior

Model of optimizing agent

Middle Layer: Model Representation

Capturing in abstract general form preferences and searching/stopping behavior:

► $u : X \rightarrow \mathbb{R}$

→ utility for each option

► $\tau : \mathcal{L} \times \mathbb{N} \rightarrow \mathbb{R}$

→ $\tau(l, N)$ is the threshold applied if the DM has observed the ordered alternatives l in a choice set they know to be of size N .

Middle Layer: Model Representation

- If following threshold $\tau(\cdot)$ and faced with a choice list $I = d, x_1, x_2, \dots, x_{N-1}$, the DM:

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- If following threshold $\tau(\cdot)$ and faced with a choice list $I = d, x_1, x_2, \dots, x_{N-1}$, the DM:

1. Looks at the first element (d)
2. Compares $\tau(d, N - 1)$ to $u(d)$

if $u(d) > \tau(d, N - 1)$

stop and choose d

otherwise continue to search

Middle Layer: Model Representation

- If following threshold $\tau(\cdot)$ and faced with a choice list $I = d, x_1, x_2, \dots, x_{N-1}$, the DM:

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3. If search continues compare $\max\{u(d), u(x_1)\}$ to $\tau(\{d, x_1\}, N - 1)$

if $\max\{u(d), u(x_1)\} > \tau(\{d, x\}, N - 1)$

stop and choose $\max\{u(d), u(x_1)\}$

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if $\max\{u(d), u(x_1)\} > \tau(\{d, x\}, N - 1)$

stop and choose $\max\{u(d), u(x_1)\}$

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4. And so on

Middle Layer: Model Representation

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Therefore:

- ▶ A *distribution* over thresholds $T \in \Delta\{\tau(\cdot)\}$ leads to an experimental data set, a distribution over triples (x, y, I)

$$\rho(x, y, I)$$

Middle Layer: Model Representation

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$$\rho(x, y, I)$$

A ***model*** is a pair $\{u, T\}$, which leads to a ρ when applied to ordered choice sets.

Theory Overview

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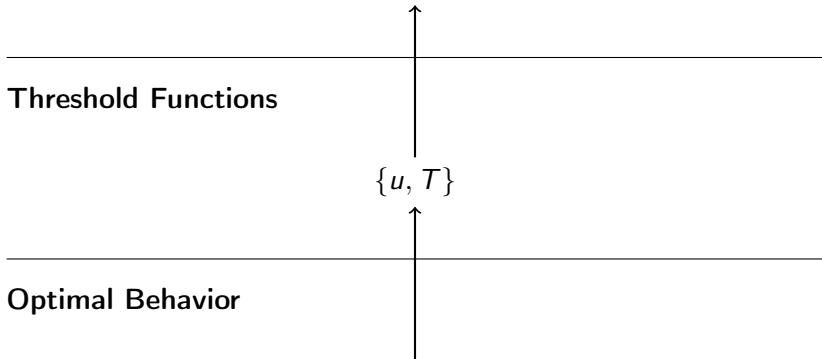
$$p(a, A), \rho(x, y, l)$$

Threshold Functions

$$\{u, T\}$$

Optimal Behavior

Model of optimizing agent



Theory Overview

Data

$p(a, A)$, $\rho(x, y, l)$: No choice overload, particular search behavior

Threshold Functions

$\{u, T\}$: Support of T with constant thresholds only

Optimal Behavior

Standard Sequential Search

Generating Choice Overload

Definition

Say that a model $\{u, T\}$ **can generate choice overload** if there are choice sets $A \subset B$ and distributions over orderings on them π_A and π_B such that the resulting experimental dataset ρ is such that

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Say that a model $\{u, T\}$ **can generate choice overload** if there are choice sets $A \subset B$ and distributions over orderings on them π_A and π_B such that the resulting experimental dataset ρ is such that

$$p(d, B) = \sum_{l \in \mathcal{L}_B} \pi_B(l) \sum_{y \in B} \rho(d, y, l) > \sum_{l \in \mathcal{L}_A} \pi_A(l) \sum_{y \in A} \rho(d, y, l) = p(d, A)$$

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Definition

Say that a model $\{u, T\}$ **can generate choice overload with *uniform search*** if the previous holds but imposing $\pi_A(l) = \frac{1}{|\mathcal{L}_A|}$.
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- ▶ Excludes cases with choice overload happening due to specific orderings
- ▶ Moving on, “can generate choice overload” means with uniform search
- ▶ “Can” generate choice overload is a feature: *set composition matters*

Theory Overview

Data

$$p(a, A), \rho(x, y, l)$$

Threshold Functions

$$\{u, T\}$$

Optimal Behavior

Search-based

Learning Increasing Costs

Classic

Decision Avoidance
Contextual Inference

Sequential Search: non-constant thresholds

$$V_n(y_n) = \max \left\{ u(y_n), -\textcolor{red}{k} + \int V_{n+1}(y_{n+1}(x)) d\textcolor{blue}{F}(x) \right\}$$

Making changes to:

- ▶ Cost function.
- ▶ (Beliefs about the) distribution of values.

Leads to non-constant thresholds:

- ▶ History dependent: what options have been seen.
- ▶ Period dependent: number of options searched.

Sequential Search with Learning

- ▶ For history-dependent thresholds, we introduce learning.
- ▶ Why learning?
 - Option values are often correlated.
 - Values of seen options are informative about the probable values of unseen options.

Sequential Search with Learning

- ▶ Choice set is drawn from one of two distributions: $F(\cdot|\overline{\mu})$ or $F(\cdot|\underline{\mu})$
- ▶ $F(\cdot|\overline{\mu})$ and $F(\cdot|\underline{\mu})$ are absolutely continuous, with respective pdfs $f(\cdot|\overline{\mu})$ and $f(\cdot|\underline{\mu})$ with Monotone Likelihood Ratio, with same support $[a, b] \subset \mathbb{R}$
- ▶ DM has belief $\mu \in [0, 1]$ over from which distribution choice set was drawn.

Sequential Search with Learning

Bellman equation:

$$V_n(y, \mu) = \max \left\{ y, -k + \int_a^y V_{n-1}(y, \mu(x)) dF(x|\mu) + \int_y^b V_{n-1}(x, \mu(x)) dF(x|\mu) \right\}$$

Sequential Search with Learning

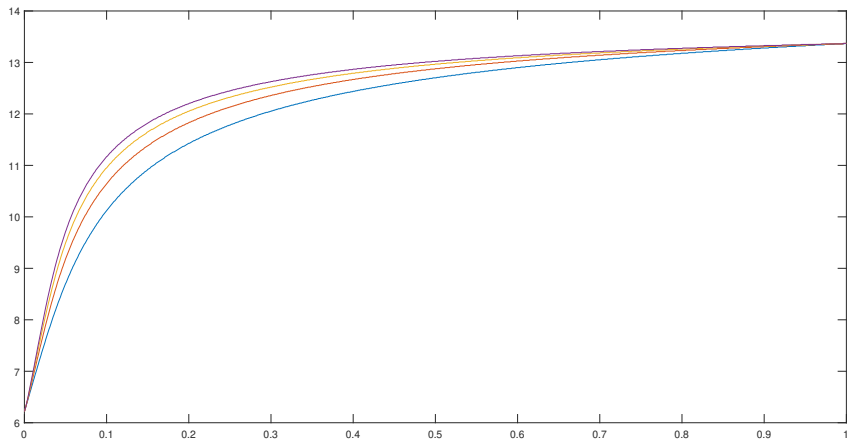
Bellman equation:

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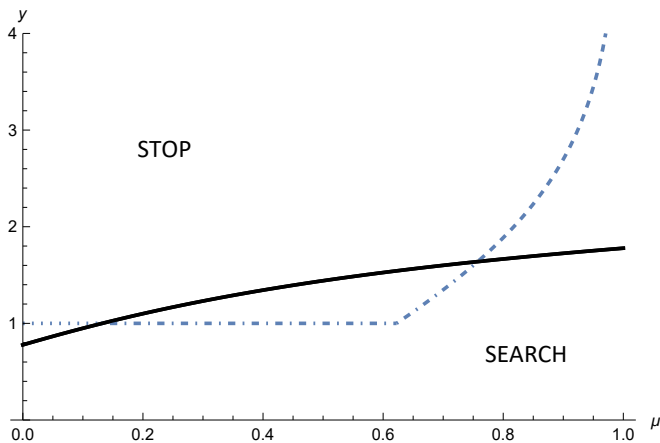
Solution: $\tau(\mu, n) = \tau(l, N)$, $n = N - \text{size}(l)$

- Increasing in beliefs: $\tau(\mu, n) < \tau(\mu', n)$, $\mu < \mu'$
- Decreasing in remaining options: $\tau(\mu, n-1) < \tau(\mu, n)$

Numerical Example: Thresholds $\tau(\mu, n)$



Looking ahead to next period



Sequential Search with Learning

Putting both beliefs and items remaining in a single dimension - *minimum cost*:

- ▶ Given a choice set A and an ordering I on it, for each $x \in A$, $m_I(x)$ is the *minimum cost* at which search stops before or at x when following the threshold induced by that cost.

Sequential Search with Learning

Putting both beliefs and items remaining in a single dimension - *minimum cost*:

- ▶ Given a choice set A and an ordering I on it, for each $x \in A$, $m_I(x)$ is the *minimum cost* at which search stops before or at x when following the threshold induced by that cost.
- ▶ Only those with $k < m_I(x)$ continue searching after x

Sequential Search with Learning

Putting both beliefs and items remaining in a single dimension - *minimum cost*:

- ▶ Given a choice set A and an ordering I on it, for each $x \in A$, $m_I(x)$ is the *minimum cost* at which search stops before or at x when following the threshold induced by that cost.
- ▶ Only those with $k < m_I(x)$ continue searching after x
- ▶ Minimum cost is all that matters to determine whether a better-than-default option is missed and whether search has stopped at a certain point.

Increasing costs: period-dependent thresholds

$$V_n(y_n) = \max \left\{ y_n, -k(n) + \int V_{n+1}(y_{n+1}(x)) dF(x) \right\}$$

For period dependent (but history independent) thresholds, choice overload can happen with:

- Decreasing search-length dependent threshold:

$$\tau(n) = \tau(l, N), n = \text{size}(l) \text{ s.t. } \tau(n+1) < \tau(n)$$

- “Fatigue”.
- As seen in Brown, Flynn and Schotter (2011).
- Optimal under cost function that increases with number of options searched: $k(n) < k(n+1)$

Classic Choice Overload

Recall what we call 'classic' choice overload: decision avoidance and contextual inference.

- ▶ Decision Avoidance

- Engaging with large choice sets is aversive
- Threshold decreasing in choice set size

- ▶ Contextual Inference

- Agent can tell that if choice set is large, distribution F is bad
- *Before* searching anything
- Leads to a lower threshold for large choice sets

In both cases, search simply doesn't start in large choice sets.

Classic Choice Overload

To capture decision avoidance, we consider thresholds decreasing in the number of alternatives left to search:

- ▶ A *decision avoidance* threshold function is such that

$$\tau(I, N) = \tau^d(N - |I|), \text{ strictly decreasing in its argument}$$

Classic Choice Overload

For contextual inference:

$$V_n(y) = \max \left\{ y, -k + \int_a^y V_{n-1}(y) dF(x|N) + \int_y^b V_{n-1}(x) dF(x|N) \right\}$$

in which N is the size of the choice set, and $F(\cdot|N)$ FOSD $F(\cdot|M)$ if $N < M$.

Classic Choice Overload

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in which N is the size of the choice set, and $F(\cdot|N)$ FOSD $F(\cdot|M)$ if $N < M$.

► Solution: threshold function such that

$$\tau(l, N) = \tau_k^l(N), \text{ strictly decreasing in its argument}$$

Classic Choice Overload

- ▶ In both cases, better-than-default options are missed because search doesn't start.
- ▶ And choosing or not choosing the default option is unaffected by list order.

Summary

	Model		
	Learning	Inc. costs	Classic
Overload?	Yes	Yes	Yes
Search Pattern	Start/Stop	Start/Stop	Don't start
Determinants of stopping	Max Val Items Rem. Beliefs	Max.Val # Searched	Max. Val

Theory to Experiment

Okay, what to test for? (Besides choice overload itself)

Theory to Experiment

Okay, what to test for? (Besides choice overload itself)

- ▶ First, separate between *classic* and *search-based* choice overload.
- ▶ Second, separate search-based between *increasing costs* and *learning*

Theory to Experiment

Search-based vs. Classic

Theory to Experiment

Search-based vs. Classic

- ▶ In classic models, good options missed because search doesn't start
- ▶ Default choice affected only by choice set size, not ordering
- ▶ In particular, not the position of the first better-than-default option

Theory to Experiment

Search-based vs. Classic

- ▶ In classic models, good options missed because search doesn't start
- ▶ Default choice affected only by choice set size, not ordering
- ▶ In particular, not the position of the first better-than-default option
- ▶ So need to generate randomization in the position of the first better-than-default option

Theory to Experiment

Search-based: Increasing costs vs. Learning

Theory to Experiment

Search-based: Increasing costs vs. Learning

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- ▶ In learning, the value of the options along the way matter

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Search-based: Increasing costs vs. Learning

- ▶ In increasing costs, it's *only* the position of the first good option that affects default choice
- ▶ In learning, the value of the options along the way matter

Need to differentiate between the two.

Theory to Experiment

We show for increasing costs:

- ▶ If there's a better-than-default alternative in the choice set, **whether default is chosen depends only on the position in the list of the first better-than-default alternative.**

Theory to Experiment

We show for increasing costs:

- ▶ If there's a better-than-default alternative in the choice set, **whether default is chosen depends only on the position in the list of the first better-than-default alternative.**
- ▶ **Whether search has stopped by a given point depends only on the number of alternatives searched and the maximum utility seen thus far.**

Theory to Experiment

We show for learning:

- ▶ When a list l a better-than-default alternative, **the probability of choosing the default depends only on $m_l(x)$** , with x the element immediately preceding the first above-default element.

Theory to Experiment

We show for learning:

- ▶ When a list I a better-than-default alternative, **the probability of choosing the default depends only on $m_I(x)$** , with x the element immediately preceding the first above-default element.
- ▶ For any list I and alternative x in I , **the probability that search has stopped at or before x depends only on $m_I(x)$** .

Introduction

Theory

- Standard Sequential Search

- Overview

- Model Representation

- Optimal Search

Experiment

Results

Experiment: task

Experimental task had to fulfill a few objectives:

- ▶ Options with clear value, but value cognitively costly to process
- ▶ Control of search order
- ▶ Make observable decision to stop searching
- ▶ Control beliefs about generating distributions

Experiment: task

- ▶ We use a task similar to the one used in Caplin, Dean and Martin (2011) and Dean, Ravindran and Stoye (WP).
- ▶ Subjects are presented with sets of experimental points (monetary values) displayed as verbal mathematical operations.
- ▶ We modify the task so that non-default options start out covered and have to be uncovered before they can be seen.
 - Options are uncovered in pre-specified order.

Experiment: task

six

Uncover

Choose "six"

Experiment: task

six

four plus zero plus zero minus four

five plus one minus three minus three

six plus two minus three minus five

Uncover

Choose "six"

Experiment: task

six

four plus zero plus zero minus four

five plus one minus three minus three

six plus two minus three minus five

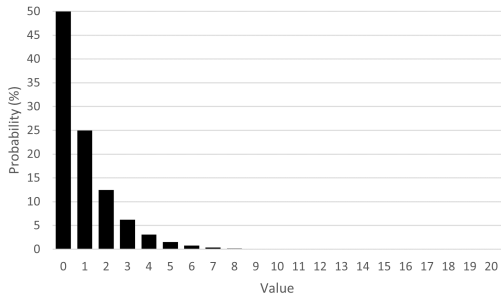
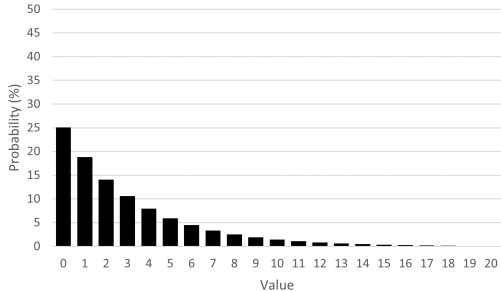
Uncover

Choose "five plus one minus three minus three"

Experiment: setup

- ▶ Options drawn from one of two distributions, randomized each round
 - Allows for learning as subject searches
 - We can calculate Bayesian beliefs at each point
 - Also ran a no-learning control
- ▶ We use geometric distributions truncated at 20 (1 point = \$0.50).
- ▶ Default is worth 6 points.
- ▶ Subjects make 10 choices. One is randomly selected for reward.
- ▶ Subjects face random sets (plus the default) of size 1 (4), 10 (2), 15 (2) and 20 (2).

Experiment: setup



Experiment: setup

- ▶ Extensive instructions with practice in drawing from the distributions to understand them and how choice sets were being constructed
- ▶ 5 comprehension questions, subjects who failed the same question twice were routed out
- ▶ 307 subjects on Prolific (314 on no learning control)

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- ▶ Question 1: Do we find choice overload?

Results

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- ▶ Yes!

Set Size	% Default Choice	<i>N</i>
2	9	53
11	38	211
16	41	273
21	44	274

- ▶ Looking at sets in which there was a better option than default.

Results

OLS regression of default choice on set size dummies:

Default Choice	
S=11	0.285*** (0.050)
S=16	0.316*** (0.050)
S=21	0.347*** (0.052)
Constant	0.094*** (0.041)
R^2	0.028
N	811

Results

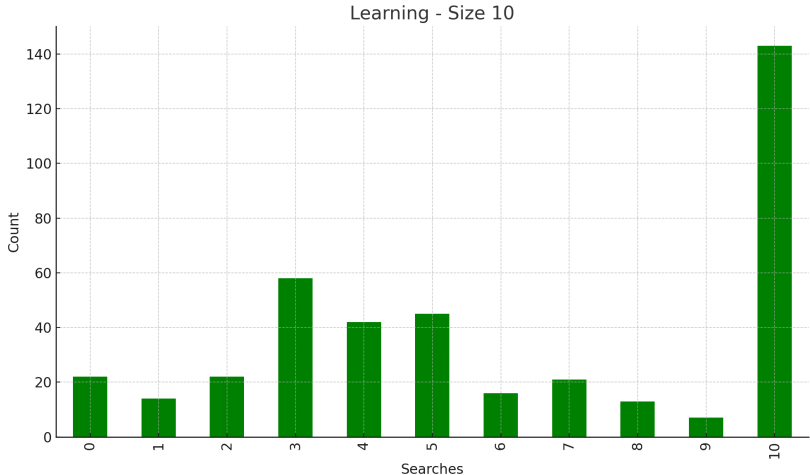
- Question 2: Is the increase in missing good options all due to never searching?

Size	Choose Default	No search	<i>N</i>
2	9%	2%	53
11	38%	7%	211
16	41%	5%	273
21	44%	5%	274
$s > 2$ minus $s = 2$	32%	4%	

Results

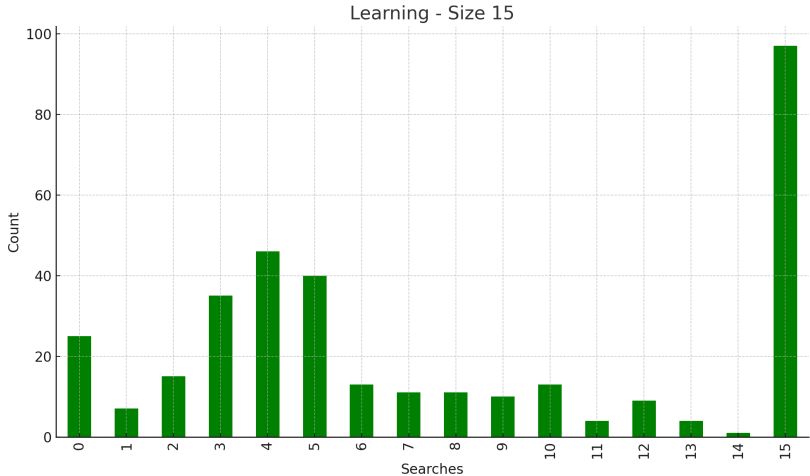
- ▶ Question 2.5: Is everyone standard or classic?
- ▶ Can look at sets *without* an option better than default.
- ▶ Under standard/classic CO models: search behavior completely on the extremes, i.e., no search or full search

Results



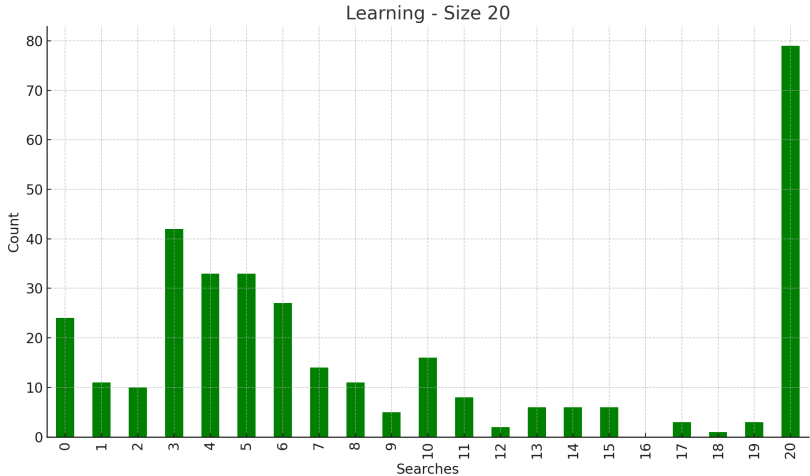
- Sets without better-than-default options
- Search distribution for size 10: 59% non-extreme

Results



- Sets without better-than-default options
- Search distribution for size 15: 64% non-extreme

Results

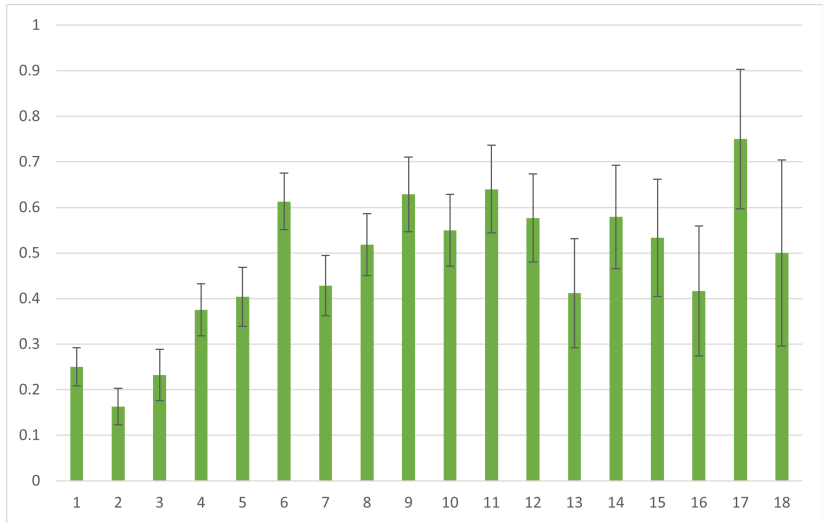


- Sets without better-than-default options
- Search distribution for size 20: 70% non-extreme

Results

- ▶ Question 3: Does search order affect choice?
 - True in search models, not in classic models of overload
- ▶ Here we can use random variation in the position of the 1st above-default option in the search sequence
- ▶ Are people more likely to choose the default when better options occur later?

Results



- Probability of default choice by position of first better option
- Significant at $<0.1\%$, robust to separate set sizes

Results

- ▶ Choice deferral may happen more often when first good option occurs later because
 - Higher search costs
 - Lower beliefs
- ▶ This leads to:
- ▶ Question 4: Does learning play a role in default choice?
 - Can fix the position of the first above default option
 - According to increasing costs: values before then should have no effect
- ▶ Again test this using random variation in search sequences

Results

- ▶ For each list position $k = 1, \dots, 15$ look at sets in which the first above default option occurs after k
- ▶ Regress

$$chose_default_{i,j} = \alpha + \beta min_cost_{i,j}^k + \varepsilon_{i,j}$$

- ▶ Where
 - i subject, j round
 - $min_cost_{i,j}^k$: lowest cost such that an optimal decision maker would stop searching at or before position k in set (i, j)
- ▶ Include also regression using beliefs

Results

Min. Beliefs				Min. Costs		
k	Coeff	s.e.	N	Coeff	s.e.	N
1	-0.14	0.10	650	0.61	0.67	650
2	-0.37*	0.20	564	-0.67	0.48	564
3	-0.51	0.32	508	-0.59	0.40	508
4	-0.61	0.45	436	-0.44	0.39	436
5	-0.35	0.60	379	0.00	0.38	379
6	-0.49	0.80	317	0.04	0.39	317
7	-1.37	1.06	261	-0.29	0.38	261
8	-2.92**	1.22	207	-0.59	0.39	207
9	-3.58**	1.50	172	-0.84**	0.40	172
10	-3.85*	1.97	132	-1.14***	0.43	132
11	-5.22**	2.32	107	-1.65***	0.46	107
12	-8.99***	2.50	81	-1.78***	0.51	81
13	-8.10*	4.50	64	-1.66***	0.54	64
14	-11.37*	6.07	45	-1.98***	0.58	45
15	-11.03	6.82	30	-1.61**	0.72	30

Results

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- ▶ Question 5: Which factors play a role in the decision to stop searching?
- ▶ Method: Run a 'behavioral' regression
 - Observation is behavior at list item k in list (i, j)
 - Regress whether or not search has stopped by k

$$stop_{i,j}^k = \alpha_i + \beta_1 above_{i,j}^k + \beta_2 k + \beta_3 min_cost_{i,j}^k + \varepsilon_{i,j}^k$$

- ▶ Where
 - $min_cost_{i,j}^k$ minimum cost at position k for set (i, j)
 - k is number of items seen
 - $above_{i,j}^k$ dummy for whether best thing seen is better than default
- ▶ Regression with beliefs and items remaining also

Results

	Searched	Stopped
	(1)	(2)
Min. Cost	-0.713*** (0.072)	
Min. Beliefs		-0.397*** (0.041)
k	0.031*** (0.001)	0.029*** (0.002)
Items Remaining		-0.007*** (0.002)
Above	0.060*** (0.017)	0.084*** (0.017)
Const	0.329*** (0.018)	0.322*** (0.026)
R^2	0.542	0.537
N	24,182	24,182
Subject f.e.	Yes	Yes

Conclusion

- ▶ Theoretically-based experimental exploration of search mechanisms of choice overload.
- ▶ General family of models capturing optimal solutions of sequential search models.
 - Captures effects of limited information processing.
- ▶ Experiment designed to test these different models.
- ▶ Results show search channel is important
 - Highlight the role of learning
- ▶ Implications for Economic Design